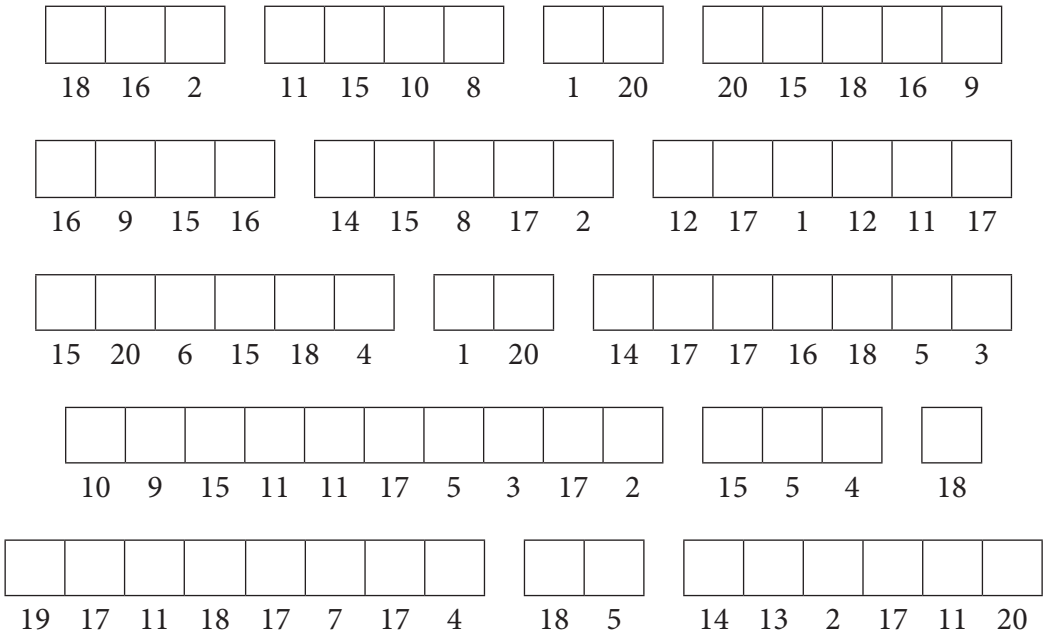
Name: _____

WORKSHEET

Sketching quadratic functions



Muhammad Ali

Rule

For a quadratic function, $f(x) = ax^2 + bx + c$:

- the ***y*-intercept** can be found by letting $x = 0$
- the **turning point** can be found at $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$
- the ***x*-intercepts** can be found using $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, if $b^2 - 4ac \geq 0$.

If $a > 0$, the function has a **minimum** at the turning point.
If $a < 0$, the function has a **maximum** at the turning point.

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Questions

Given $f(x) = x^2 + 4x - 5$:

1 What is the y-intercept?

2 What are the x-intercepts?

3 Where is the turning point?

4 Sketch the function.

Given $f(x) = x^2 + 7x + 12$:

5 What is the y-intercept?

6 What are the x-intercepts?

7 Where is the turning point?

8 Sketch the function.

Given $f(x) = -5x^2 + 10x - 3$:

9 What is the y-intercept?

10 What are the x-intercepts?

11 Where is the turning point?

12 Sketch the function.

Given $f(x) = 2x^2 + 12x + 9$:

13 What is the y-intercept?

14 What are the x-intercepts?

15 Where is the turning point?

16 Sketch the function.

Given $f(x) = 12x - 3x^2 - 7$:

17 What is the y-intercept?

18 What are the x-intercepts?

19 Where is the turning point?

20 Sketch the function.

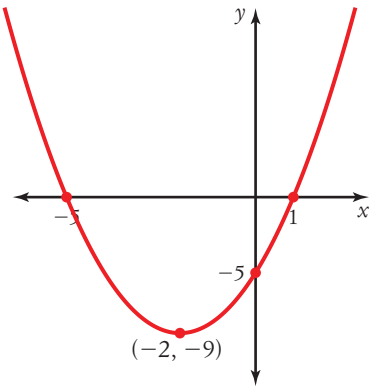
Solutions

A $(-3, -9)$

B $(2, 5)$

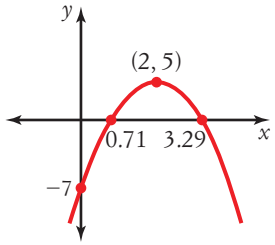
C $1 - \frac{\sqrt{40}}{10}, 1 + \frac{\sqrt{40}}{10}$

D



E -7

F

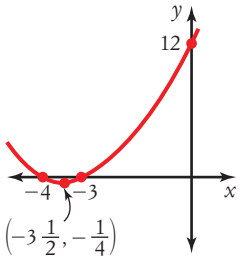


G $(-2, -9)$

H -3

I $2 - \frac{\sqrt{60}}{6}, 2 + \frac{\sqrt{60}}{6}$

K



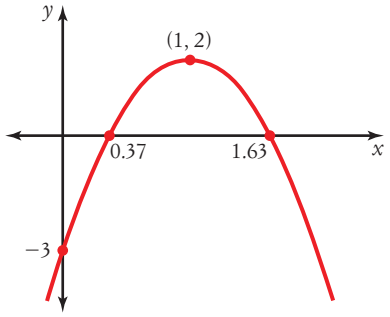
L $(1, 2)$

M $-3 - \frac{\sqrt{72}}{4}, -3 + \frac{\sqrt{72}}{4}$

N 12

O -5

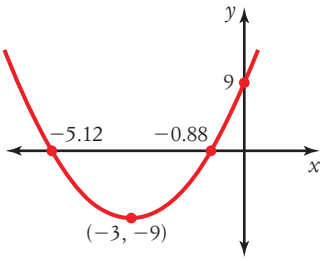
P



R $-3, -4$

S $-5, 1$

T



V $\left(-3\frac{1}{2}, -\frac{1}{4}\right)$

Y 9

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